When Overall AUC Hides Miscalibration: Evaluating Knowledge Tracing on Rarely Practiced Knowledge Components

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Abstract*—Knowledge tracing (KT) models are usually ranked by the area under the receiver operating characteristic curve (AUC) over all responses, which can hide poor calibration on knowledge components (KCs) that learners rarely practice. We fix one protocol on ASSISTments 2012: the validation-AUC checkpoint, non-overlapping test windows of at most 200 responses, 15 equal-width bins, frequency groups, and four unique learner partitions. For deep knowledge tracing (DKT), the mean expected calibration error (ECE) was 0.0982 on sparse KCs and 0.0089 on dense KCs, a difference of 0.0893. When dense rows were subsampled to the sparse sample size, the gap was 0.0400, and its learner-cluster 95% interval on one partition was 0.0170 to 0.0821, although the conservative small-sample bias reference averaged −0.0081. A causally masked Transformer showed the same direction (difference 0.1041). In a DKT checkpoint-by-window grid, the validation-AUC checkpoint was associated with a sparse–dense difference lower by 0.0181 to 0.0774 than the final checkpoint, whereas changing the test window shifted it by −0.0131 to +0.0068. The evidence comes from one dataset with a mean of 398 sparse responses per partition; Junyi Academy had no sparse group. Reports should give frequency-stratified calibration, group sizes, and the checkpoint rule alongside overall AUC.

***Keywords*—knowledge tracing, calibration, knowledge components, adaptive learning, learning analytics, evaluation protocol**

I. INTRODUCTION

Adaptive practice platforms and intelligent tutoring systems use knowledge tracing (KT) to estimate how well a learner has mastered a skill and to decide whether to give more practice, assign remediation, or move on [1]–[3]. Such decisions often rest on the predicted probability p that the learner will answer the next item on the skill correctly: once p reaches a threshold τ, further practice on that skill is withheld. A rule of this kind takes p at face value. A prediction of 0.7 should mean that about seven in ten comparable responses are correct; in other words, the model must be well calibrated.

KT models, however, are usually compared by the area under the receiver operating characteristic curve (AUC) and accuracy computed over all test responses [4], [5]. AUC measures ranking, that is, whether correct responses tend to receive higher probabilities than incorrect ones; calibration is a separate property. The difference matters most for knowledge components (KCs) that learners rarely practice. In the training data used here, 18.9% of the KCs in ASSISTments 2012 and 21.5% of the XES3G5M KCs present in training had fewer than 100 training rows (Table 7), whereas frequently practiced KCs supplied most of the rows. Pooled metrics therefore mainly reflect frequently practiced KCs, and a model can rank responses well overall while assigning overconfident probabilities to rarely practiced KCs. Under a fixed threshold, such overconfidence appears as an advance decision followed by an incorrect next response; we call the proportion of these decisions the false-advance rate (FAR).

Earlier work has examined parts of this problem. Benchmark studies have standardized KT evaluation and documented label leakage [5], [6], and calibration, uncertainty, cold start, and mastery thresholds have been studied in their own right (Section II). This study combines these elements into one evaluation protocol for standard KT models; it does not propose a new model.

The study addresses three research questions. Only ASSISTments 2012, under the primary protocol (Section III.C), is used for the primary answers. Junyi Academy is a post-hoc check. XES3G5M contributes distribution and estimability, not model scores.

RQ1: Under the primary protocol, how does expected calibration error (ECE) differ between sparse and dense knowledge components on ASSISTments 2012, once the bias reference and a size-matched comparison are reported separately?

RQ2: When is a sparse-versus-dense contrast estimable, and when must model-based results be withheld?

RQ3: On ASSISTments 2012, how does the deep knowledge tracing (DKT) contrast change when the checkpoint and the test window are varied one at a time?

For RQ1, DKT and a causally masked Transformer are the primary models. Across four unique partitions their sparse ECE exceeds their dense ECE. The size-matched gap is positive, and the fold-0 interval for that gap excludes 0. ΔECE* does not stay positive on every partition, so the bias reference and the matched interval are not the same statement. Section IV gives the partition means. For RQ2, Junyi Academy has no sparse group. XES3G5M has a low-frequency tail in its KC-level file, but check L8 failed and no corrected model was trained, so it is a descriptive audit only. For RQ3, in the DKT grid of Table A2 the validation-AUC checkpoint was associated with a ΔECE lower by 0.0181 to 0.0774 than the final checkpoint at a fixed test window, in every reported run, whereas changing the window at a fixed checkpoint changed ΔECE by −0.0131 to +0.0068. Two runs on one learner partition differ by up to

0.0268, so these are checkpoint-associated differences for one model and dataset, not a general causal estimate.

The paper makes three contributions. First, it specifies a protocol that groups knowledge components by training frequency, labels sample support, defines ΔECE* as a bias reference rather than a test, and states eight leakage checks. Second, it reports ASSISTments DKT and the causally masked Transformer (T-KT masked) under that protocol, with the raw contrast, the bias reference, and the size-matched gap kept apart. Third, it states the limits of the evidence: Junyi Academy is post-hoc and has no sparse group, XES3G5M is a descriptive audit only, and a change of both checkpoint and test window is not a single-cause result. Unmasked T-KT is not used as a finding.

II. LITERATURE REVIEW

A. Knowledge tracing models and benchmarks

KT models a learner's changing proficiency from past interactions in order to predict the next response [3]. Bayesian knowledge tracing (BKT) tracks a latent mastery state for each skill [1]. Item response theory (IRT), in its one-parameter Rasch form [7], relates the probability of a correct answer to learner ability and item difficulty. DKT replaced hand-specified learning dynamics with a recurrent network and established next-response prediction as the standard task [2]. Attention-based successors, such as context-aware attentive knowledge tracing [8] and simpleKT [4], keep that task but replace recurrence with self-attention. The best model depends on the data: in a comparison on nine real-world datasets, logistic regression with a suitable feature set led on datasets of moderate size, whereas DKT led on the largest ones [9]. Uncertainty-aware models such as Uncertainty-Aware Knowledge Tracing (UKT) represent knowledge states as distributions [10].

B. Evaluation rigor and leakage

Benchmark libraries such as pyKT have standardized preprocessing and model comparison and have shown that evaluation choices can distort reported discrimination [5]. One such choice concerns questions tagged with several KCs. When such a question is expanded into one row per KC and each row is scored separately, a model can use the answer to the same question from the previous row. In pyKT's experiments, this raised AUC by 8.38 and 13.09 percentage points on average on ASSISTments 2009 and Algebra 2005, respectively [5]. Badran and Preisach analyze the same form of label leakage [6], and our protocol defines this route as check L8. The protocol applies the same discipline to frequency groups and calibration: each KC's group and descriptors are computed from training data only, and no cut point or threshold is tuned on test data. Graph-based KT [11] and contrastive learning for KT [12] form related research lines that focus on new model architectures.

C. Four meanings of sparsity

The word "sparse" has at least four meanings in KT research. (a) Sparse attention: sparseKT applies *k*-sparse attention so that only a subset of past interactions receives nonzero weight [13]. (b) Sparse KC frequency: a KC with few observations in the training data. (c) New-student cold start: prediction for learners with little or no history at test time [14]. (d) Concept-level cold start: a KC that never appears in the training data, whether or not the learner is new. These problems are distinct: sparse attention can be applied to a frequently practiced skill, and a learner seen in training can meet a KC absent from it. This study addresses meanings (b) and (d) through training-frequency groups that include a strict cold-start group ($f_{\text{train}} = 0$).

D. Calibration and educational decisions

When predicted probabilities drive educational decisions, ranking alone is an insufficient evaluation [15]. Calibration asks whether a predicted probability matches the observed frequency of correct responses. ECE summarizes the mismatch over probability bins [16], and modern neural networks can be poorly calibrated even when they discriminate well [17]. The Brier score [18], Murphy's decomposition into uncertainty, resolution, and reliability [19], and the calibration–refinement view [20] separate miscalibration from the difficulty of the task, while reliability diagrams display the same comparison graphically.

Within KT, post-hoc per-item bias correction has been used to recover discrimination in a trained model [21], and selective prediction defers the least certain predictions to a teacher [22]. Mastery-learning research has examined false positives at the mastery threshold [23] and predictive performance when a student starts a skill [24]. Our threshold analysis complements selective prediction: it does not abstain, but measures how often the advances made under a fixed threshold are followed by an incorrect response, separately for each frequency group.

III. MATERIALS AND METHODS

Fig. 1 outlines the evaluation workflow; the leakage checks attached to its steps are defined in Section III.F.

A. Datasets

We used three de-identified public logs mapped to a common interaction schema: ASSISTments 2012 [25], Junyi Academy [26], and XES3G5M [27]. Preprocessing removed rows with missing learner, KC, or label identifiers, binarized correctness, parsed timestamps, and kept learners with at least two interactions. Table 1 reports the counts after these filters, which therefore differ from those of the raw public files. The processed logs contain 2,657,490 (ASSISTments 2012), 16,215,567 (Junyi Academy), and 6,413,353 (XES3G5M) rows.

Table 1. Cohort statistics after preprocessing (learner-based split; test events from fold 0). For XES3G5M, interactions and test events are KC-level rows (Section III.A). Test events are all test rows of fold 0; Tables 4, 5, and A1–A5 use only the rows with a nonempty test history.

Dataset	Learners	KCs	Interactions	Test events
ASSISTments 2012	27,806	265	2.66M	534,150
Junyi Academy	71,014	1,326	16.2M	3,269,022
XES3G5M	18,066	865	6.41M	1,282,422

ASSISTments 2012 is the 2012–2013 public release with affect [25], which is also supported by the pyKT toolkit [28]; each retained row carries one skill_id. Junyi Academy is the

Kaggle Online Learning Activity dataset [26], which differs from the 2015 Junyi release. We used its ucid field, the content (exercise) identifier that the practice log assigns to each item, as the KC; it is an operational identifier rather than a curated skill tag. Practice volume per exercise is high: across the whole log, the 1,326 exercises received about 12,200 responses each on average. In learner-based fold 0, every exercise had at least 100 training responses (Fig. 2), and the sparse group was empty in all four learner-based partitions, so Junyi Academy has no sparse group under the learner-based split.

XES3G5M was loaded from the authors' KC-level release: the training/validation file and the test file were concatenated and flattened to one row per valid sequence position, with padding, invalid labels, and rows with a negative KC identifier removed. The dataset paper reports 18,066 students, 7,652 questions, 865 KCs, and 5,549,635 question-level interactions [27]. Our file has 6,413,353 rows because a question tagged with several KCs is expanded into one row per KC; the number of distinct items remains 7,652, and all XES3G5M counts, including the 1,282,422 test events of fold 0 (Table 1), are KC-level rows. If the model input for one row contains the outcome recorded at another row of the same response, the target label is visible to the model [5], [6]. Our audit of this route (check L8, Section III.F) in the code used for the XES3G5M runs found that the KC rows of a response stay adjacent and that both training and scoring condition each prediction on the KC–response tokens of earlier rows. For the second and later KC rows of a multi-KC response (863,807 rows by the release's repeat flags, 13.5% of the file; counted in artifact v4, audits/xes_padding_counts.csv), the input therefore contains the outcome being predicted, in training and at test time, so L8 fails. Because removing this route requires retraining, which this study did not do, we report no model-based results for XES3G5M; the dataset still contributes the frequency distribution and estimability conditions (Fig. 2 and Section IV.C), which do not depend on predicted values. In ASSISTments 2012 and Junyi Academy, each released row records one response with one KC identifier and no expansion step was applied, so the route does not arise.

B. Splits and partitions

The learner-based split is the primary design. For each partition, learners were shuffled with the partition seed, and 20% were held out for testing, 10% for validation, and the rest for training, so the three sets share no learners. A temporal split served as a complementary check: all interactions were sorted by timestamp, and the earliest 70%, the next 10%, and the latest 20% formed the training, validation, and test sets. It was run once (seed 42), and this paper reports from it only the temporal distributions in Fig. 2; all performance results come from the learner-based split. In both designs, KC frequencies, and hence group membership, were computed from the training set only.

Deep models were trained with seeds 42, 2024, 2025, 2026, and 2027 (folds 0–4). On ASSISTments 2012 and Junyi Academy, seeds 2025 and 2026 share one learner partition (fold 2 = fold 3), so the five runs cover four distinct partitions, and the two runs on the shared partition were averaged before averaging across partitions. Unless stated otherwise, every mean ± standard deviation reported here is therefore an equal-weight summary over four learner-based partitions.

C. Models

Table 2 lists the runs that supply the primary and sensitivity scores. DKT is a long short-term memory (LSTM) network over KC–response tokens and pads with index 0. T-KT masked is a two-layer Transformer over the same tokens, with a causal mask, no positional encoding (order enters only through the mask), and padding id 2n (n is the number of KCs). Both were trained on ASSISTments 2012 for at most 50 epochs, with patience 8, and the saved checkpoint is the one with the highest validation AUC. Test sequences are scored in non-overlapping windows of at most 200. We call this setting the primary protocol (internal ID P0-BV200-15EW): the validation-AUC checkpoint, non-overlapping test windows of at most 200 responses, and M = 15 equal-width bins (Section III.E). simpleKT is the pyKT implementation and is a sensitivity analysis, not a primary model. BKT was not scored.

The primary checkpoint is the validation-AUC checkpoint, not the final epoch. The final epoch and the full test history appear only as sensitivity cells in Table A2. That grid was pre-specified for ASSISTments DKT in an analysis plan committed on 21 October 2026 (commit 46e0208, before the first supplementary run) and included in artifact v4. Junyi DKT uses the same checkpoint and window rules, but it was not in the pre-specified plan and is not called pre-specified.

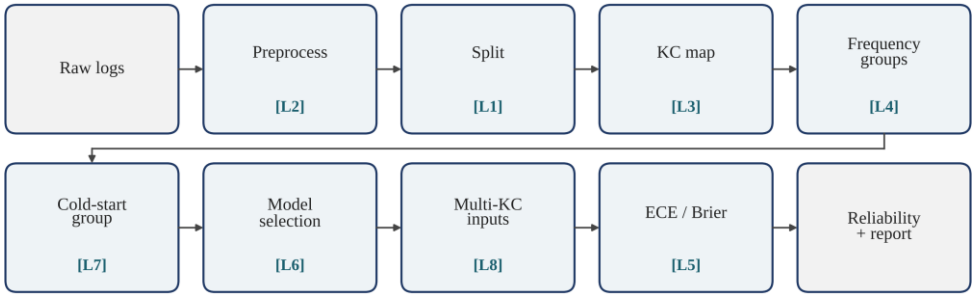


Fig. 1. Evaluation workflow. Bracketed tags give the leakage checks of Table 3 (L1–L7 numbered as in the archived audit records); they do not follow the left-to-right order. L8 applies only to datasets in which one response carries several KCs.

Table 2. Configuration of the scored neural models. Primary rows are ASSISTments DKT and T-KT masked under the primary protocol. simpleKT is sensitivity. No missing setting was filled in.

Setting	DKT	T-KT masked	simpleKT
Architecture	LSTM	Causal Transformer	pyKT simpleKT
Embedding size	64	64	256
Hidden size	128	64	256
Number of layers	1	2	2
Attention heads	none	4	8
Positional encoding	None (recurrent)	None (causal mask only)	As in pyKT
Optimizer	Adam	Adam	Adam
Learning rate	0.001	0.001	0.001
Batch size	64	64	24
Max epochs	50	50	50
Patience	8	8	8
Checkpoint rule	Best validation AUC	Best validation AUC	Best validation AUC
Test history rule	Non-overlapping chunks of 200	Non-overlapping chunks of 200	Non-overlapping chunks of 200
Split rule	Learner-disjoint; 20% test users	Same learner split	Same learner split
Provenance	Commit 4ff6683; baseline_runner.py	Commit 4ff6683; protocol_v2.py	Commit 4ff6683; p0_retrain_official_simplekt.py

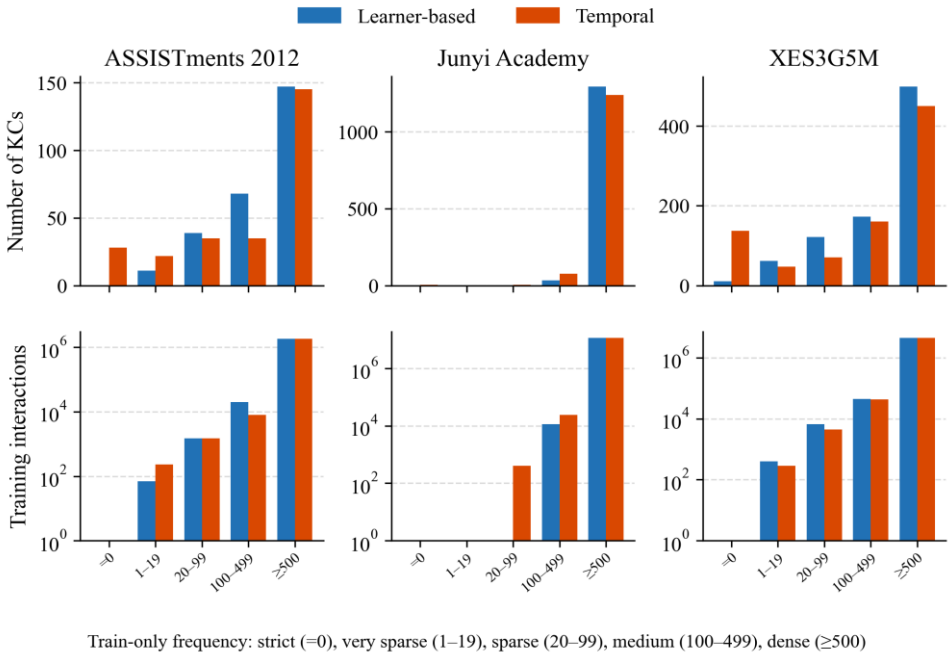


Fig. 2. Number of KCs (top) and number of training responses (bottom; sum of f_{train} in fold 0, logarithmic axis) in each training-frequency group under the learner-based and temporal splits.

D. Notation, frequency groups, and sample support

Each row of a processed log is a tuple (u, q, c, y). Learner u answered item q, c is the KC attached to the row, and $y \in \{0, 1\}$ records whether the response was correct. A partition divides the learners into disjoint training, validation, and test sets, and D_{test} denotes the test rows with a valid exported prediction. The training frequency $f_{train}(c)$ of KC c is its number of training rows. A group map g assigns each KC to one of five groups: strict cold start ($f_{train} = 0$), very sparse ($0 < f_{train} < 20$), sparse ($20 \le f_{train} < 100$), medium ($100 \le f_{train} < 500$), or dense ($f_{train} \ge 500$). Fig. 2 shows how many KCs and training responses fall in each group. The test rows of group k are

$$T_k = \{i \in D_{test} : g(c_i) = k\}, \quad N_k = |T_k| \qquad (1)$$

and every group-level metric is computed on T_k alone. N_k is counted in each partition; the result tables report its mean over the four partitions and assign support labels from this mean, whereas the leakage audit (Section III.F) labels each partition. Analyses that use another subset, such as the threshold analysis, state their own N. A group is labeled Reliable (R) if $N \ge 1{,}000$, Limited (L) if $100 \le N < 1{,}000$, and Insufficient (I) if $N < 100$. These cut points are conventions of this study that flag sample support; they are neither confidence intervals nor hypothesis tests. A metric can be computed for any nonempty group but is interpreted only for Reliable or Limited groups; Insufficient groups are reported descriptively. Likewise, a sparse-versus-dense contrast is computable whenever both groups are nonempty, and we treat it as estimable if and only if (E1) some KCs have $f_{train}(c) < 100$ in the training split and (E2) the sparse and the dense group are both at least Limited; E2 presupposes E1. In each partition, the protocol thus (i) assigns groups from training data, (ii) labels their support, (iii) checks E1 and E2, (iv) reports discrimination and calibration by group, with the bias bound of Section III.E, and (v) may apply one fixed threshold to estimate decision errors (Section III.G).

E. Calibration metrics

Within a group, let $p_i \in [0, 1]$ be the predicted probability of a correct response and y_i the observed outcome. With M = 15 equal-width bins, ECE is

$$\mathrm{ECE} = \Sigma_m \, (n_m / N) \, |acc_m - conf_m| \tag{2}$$

where N is the number of rows in the group, n_m the number in bin m, and acc_m and $conf_m$ the observed proportion correct and the mean predicted probability in that bin. The Brier score is

$$\mathrm{Brier} = (1/N) \, \Sigma_i \, (p_i - y_i)^2 \tag{3}$$

and its binned Murphy decomposition [19] is

$$\mathrm{Brier} \approx \mathrm{UNC} - \mathrm{RES} + \mathrm{REL} \tag{4}$$

with uncertainty UNC = $\bar{y}(1 - \bar{y})$, resolution RES = $(1/N) \, \Sigma_m \, n_m(acc_m - \bar{y})^2$, and reliability REL = $(1/N) \, \Sigma_m \, n_m(conf_m - acc_m)^2$, where $\bar{y}$ is the proportion correct in the group. Because the components are computed on binned probabilities, they need not sum exactly to the Brier score. Lower ECE and REL indicate better calibration, and neither replaces AUC. The main calibration contrast is

$$\Delta\mathrm{ECE} = \mathrm{ECE}_{\mathrm{sparse}} - \mathrm{ECE}_{\mathrm{dense}} \tag{5}$$

where each term is the mean over the four learner-based partitions. Partition-level standard deviations describe variation across partitions and are not confidence intervals for ΔECE. Binned ECE also depends on the number of bins and is biased upward when bins contain few responses [29], [30], so a small sparse group can by itself make ΔECE positive. To bound this bias, suppose the outcomes in a group were independent draws with the predicted probabilities. Then $acc_m - conf_m$ has mean zero and variance at most $1/(4n_m)$, so $\mathrm{E}|acc_m - conf_m| \leq 1/(2n_m^{1/2})$, and the Cauchy–Schwarz inequality gives

$$\mathrm{E}[\mathrm{ECE}] \leq \mathrm{ECE}_0(N) = \tfrac{1}{2} \, (M/N)^{1/2} \tag{6}$$

For M = 15, ECE_0 is 0.095 at N = 415 and 0.003 at N = 523,971. By the same argument, the expected ECE of any model exceeds its value at the bins' expected accuracies by at most ECE_0, so the conservative contrast

$$\Delta\mathrm{ECE}^* = \mathrm{ECE}_{\mathrm{sparse}} - \mathrm{ECE}_0(N_{\mathrm{sparse}}) - \mathrm{ECE}_{\mathrm{dense}} \tag{7}$$

is positive only if the sparse–dense difference exceeds this upper bound on expected small-sample bias. ECE0 is computed from the focal N of that partition. Fold 2 and fold 3 are averaged first, and the reported ΔECE* is the mean of the four unique partitions. It is not the result of subtracting ECE0 once from the mean ΔECE. ΔECE* is not a confidence interval, not a hypothesis test, and not a p-value. The size-matched gap, and the learner-cluster interval computed on fold 0 only, are different quantities.

The size-matched gap is computed on rows with a nonempty test history. Sparse rows belong to KCs with training frequency $20 \leq \mathrm{ftrain} < 100$, and dense rows belong to KCs with $\mathrm{ftrain} \geq 500$. On one partition, $\mathrm{N}_{\mathrm{sparse}}$ is the number of sparse rows and $\mathrm{ECE}_{\mathrm{sparse}}$ is their ECE under (2). The dense reference uses the learners who have at least one dense row in that same row set, in order of first appearance in the scored predictions. Draw r permutes those learners without replacement and appends their dense rows until the concatenated length is at least $\mathrm{N}_{\mathrm{sparse}}$. A draw with fewer than $\mathrm{N}_{\mathrm{sparse}}$ dense rows is invalid and is omitted from the median. Otherwise a second sample, also without replacement, retains exactly $\mathrm{N}_{\mathrm{sparse}}$ rows, so a learner whose rows cross $\mathrm{N}_{\mathrm{sparse}}$ need not contribute every row. The ECE of those retained rows, again under (2), is (8). The point estimate uses R = 2000 draws. In the published fold-0 run, all 2000 draws were valid. The first of the 15 equal-width bins in (2) contains predicted probabilities on [0, 1/15]; each later bin contains probabilities on (left, right].

$$\mathrm{ECE}_{\mathrm{dense},r}@\mathrm{N}_{\mathrm{sparse}} = \mathrm{ECE}(y^{(r)}, p^{(r)}) \tag{8}$$

$$G = \mathrm{ECE}_{\mathrm{sparse}} - \mathrm{median}_r(\mathrm{ECE}_{\mathrm{dense},r}@\mathrm{N}_{\mathrm{sparse}}) \tag{9}$$

Matching equalizes sample size only. It does not equalize ECE bin occupancy, the predicted-probability distribution, or the dependence structure.

The 95% interval is the fold-0 learner-cluster bootstrap. The seed 20261021 starts a NumPy Generator, and fold 0 is the first partition, so the seed is the start of the stream. That generator is then used in three stages. Stage 1 is the R = 2000 draws that define G. Each draw calls permutation on the dense learners, then choice of exactly $\mathrm{N}_{\mathrm{sparse}}$ rows with replace=False. Stage 2 calls choice 2000 times, with replace=False, to take exactly $\mathrm{N}_{\mathrm{sparse}}$ rows from the pooled dense rows in prediction-file order. Stage 2 does not enter G or the interval. A replay of the interval still runs stage 2, because those calls advance the generator. Stage 3 draws B = 2000 resamples. Each resample calls integers to draw as many learner indices as there are test learners with a nonempty history, on the half-open range from 0 to that count. An index that occurs more than once is stored as a separate cluster, and the sparse and dense rows are rebuilt from the copies. A replicate with no sparse rows is counted, left out of the percentiles, and left without an imputed gap. A replicate whose dense rows are all empty is left out in the same way. The published fold-0 run had 0 empty sparse replicates and 0 other invalid replicates, so the percentiles use all 2000 values. For each valid replicate the parent generator calls integers once on the half-open interval [0, 2^31 - 1), and that integer seeds the inner generator. The inner generator repeats the stage-1 draw 200 times, and the replicate stores its own G. The interval is the 2.5th and 97.5th percentiles of those values under NumPy's linear percentile. As a supplementary analysis, the same procedure was applied to fold 1, folds 2–3, and fold 4, with a generator seeded [20261021, f] for the first fold f of each partition; for folds 2–3, each replicate resamples the shared learners once, computes G on both runs with separate inner seeds, and stores their mean (Table A5).

F. Leakage checks

The protocol defines eight leakage checks. L1–L7 follow the numbering of the archived per-partition audit records, which cover only these seven. L1: training, validation, and test learners are disjoint. L2: every fitted transform uses training rows only; the row filtering of Section III.A fits no

parameters. L3: the KC map is the static mapping supplied by the platform. L4: groups are assigned by applying the fixed cut points to f_{train} only. L5: no calibration map is fitted on test data. L6: no checkpoint or setting is selected with test data. L7: the strict cold-start group is defined by $f_{train}(c) = 0$. L8: when one response is expanded into several KC rows, the outcome recorded at one row does not enter the model input used to train on or score another row of the same response.

Table 3 reports the outcomes for the four learner-based partitions: checks L1–L7 passed in every partition. The archived audit computes L1, L4, and L7 from the split files of each partition; L2, L3, L5, and L6 are properties of the pipeline code (fitted values use training rows only, the KC map is static, no calibration map is fitted, checkpoints are selected on validation data) and were recorded from the code, not recomputed per partition. simpleKT reads the same partition files and selects epochs on validation learners only (Section III.C), so L1, L2, and L6 also hold for it by construction. The KC map (L3) is the released skill_id for ASSISTments 2012, the released KC mapping for XES3G5M, and the operational ucid for Junyi Academy. Check L8 does not apply to ASSISTments 2012 and Junyi Academy, where each released row records one response with one KC and no row was expanded. For XES3G5M, it was not part of the archived audit, and our audit of the code found that it fails (Section III.A). Test data were used only to compute the reported metrics.

Table 3. Leakage checks over the four learner-based partitions (L1–L7 numbered as in the archived audit records). PASS means that the condition held in every partition; parentheses give the occupancy of the affected group (I, Insufficient). L8 was audited from the code (Section III.A).

Check	Condition	ASSISTments 2012	Junyi Academy	XES3G5M
L1 Split	Training, validation, and test learners disjoint	PASS	PASS	PASS
L2 Preprocessing	Fitted transforms use training rows only	PASS	PASS	PASS
L3 KC map	Static, platform-provided mapping	PASS	PASS	PASS
L4 Frequency groups	Cut points applied to training counts only	PASS	PASS (sparse group empty)	PASS
L5 Calibration	No calibration map fitted on test data	PASS	PASS	PASS
L6 Model selection	No checkpoint or setting selected with test data	PASS	PASS	PASS
L7 Cold-start group	$f_{train} = 0$, defined from training data	PASS (empty in 3 partitions, I in 1)	PASS (empty)	PASS (I)
L8 Multi-KC rows	Outcome of a response hidden from the input for its other KC rows	N/A (one row per response)	N/A (one row per response)	FAIL (visible in training and test)

G. Secondary analyses

Two secondary analyses complement the group-level results. The first concerns KC structure. For each KC, we computed four training-only descriptors. The difficulty proxy is 1 − mean training correctness, an observed rate and not an IRT parameter. Item support is the number of distinct items per KC, learner exposure the number of distinct training learners per KC, and the curriculum-position proxy the median normalized position of the KC in learners' sequences. We report Spearman correlations of $\log(1 + f_{train})$ with the difficulty proxy (frequency–difficulty coupling) and with the curriculum-position proxy. These descriptors use no model output.

The second is a simulated threshold analysis on the primary-protocol predictions of ASSISTments DKT and T-KT masked. With one fixed threshold $\tau = 0.7$ for all groups, the simulated system advances the learner (skips remediation) when $p \geq \tau$ and assigns remediation otherwise. The false-advance rate is

$$\mathrm{FAR} = P(y = 0 \mid p \geq \tau) \tag{10}$$

that is, the proportion of advance decisions followed by an incorrect response; FAR equals one minus the precision of the advance decision, or its false discovery rate. We also report three related quantities: the false-positive rate $\mathrm{FPR} = P(p \geq \tau \mid y = 0)$; the expected FAR implied by the model's own probabilities, $\mathrm{E[FAR]} = \mathrm{E}[1 - p \mid p \geq \tau]$; and the excess FAR, observed minus expected. The contrast $\Delta\mathrm{FAR} = \mathrm{FAR}_{sparse} - \mathrm{FAR}_{dense}$ is positive when advances on sparse KCs fail more often. FAR in (10) is computed over the N_{adv} responses with $p \geq \tau$, and FPR over the N_{inc} incorrect responses. Only rows with a nonempty test window enter. On fold 0, 95% intervals for FAR and ΔFAR come from a KC-clustered percentile bootstrap: each of B = 2,000 replicates resamples KCs with replacement within the sparse group and within the dense group, with a fresh NumPy Generator seeded with 20261021 for each model. A replicate with no advance in either group would be left out; none occurred. Across partitions, ΔFAR is summarized over the four unique partitions, with fold 2 and fold 3 averaged first. Because y records the next response and not latent mastery, FAR is a simulated proxy for decision error.

IV. RESULTS

A. Overall discrimination

Table 4 reports overall AUC and accuracy on rows with a nonempty test window. Under the primary protocol, ASSISTments DKT has AUC 0.7303 ± 0.0005 and accuracy 0.7354 ± 0.0008. T-KT masked has AUC 0.7244 ± 0.0007 and accuracy 0.7314 ± 0.0006. The mean and standard deviation (SD) use four unique partitions. Junyi DKT, under the same scoring rules but as a post-hoc check, has AUC 0.7554 ± 0.0012. simpleKT on ASSISTments, also post-hoc, has AUC 0.7711 ± 0.0013 and is not ranked against the primary models. These are discrimination scores under the primary protocol; they are not comparable with published benchmark scores that use other splits or window rules.

Table 4. Overall AUC and accuracy on nonempty test windows (mean ± SD across four unique learner partitions). N is the mean count, rounded. Fold 2 and fold 3 are one partition and are averaged before the SD. The SD is descriptive. All rows use the primary protocol. Junyi DKT and simpleKT are post-hoc sensitivity rows, not primary claims.

Dataset	Model	AUC ± SD	ACC ± SD	N
ASSISTments 2012	DKT (primary)	0.7303 ± 0.0005	0.7354 ± 0.0008	523,612
ASSISTments 2012	T-KT masked (primary)	0.7244 ± 0.0007	0.7314 ± 0.0006	523,612
Junyi Academy	DKT (post-hoc)	0.7554 ± 0.0012	0.7469 ± 0.0013	3,204,343
ASSISTments 2012	simpleKT (sensitivity)	0.7711 ± 0.0013	0.7533 ± 0.0014	523,612
XES3G5M	Any model	Not evaluated	Not evaluated	Not evaluated

B. Calibration across frequency groups

On ASSISTments 2012, under the primary protocol, sparse ECE is higher than dense ECE for both primary models (Table 5). For DKT the four-partition means are 0.0982 ± 0.0093 sparse and 0.0089 ± 0.0013 dense, so ΔECE = 0.0893 ± 0.0091. ΔECE* = −0.0081 ± 0.0122. The size-matched gap is 0.0400 ± 0.0107. On fold 0 that gap is 0.0351, and the learner-cluster 95% interval is 0.0170 to 0.0821 (N sparse = 430). ΔECE* on that fold is −0.0088. The matched gap is positive on all four partitions. ΔECE* is positive only on fold 1. The interval is for fold 0. It is not a test of the other partitions, and ΔECE* is not that interval. Table A5 gives G and the same learner-cluster interval for each unique partition; for both models every interval excludes 0, but these supplementary intervals are not combined into one significance claim. For T-KT masked the four-partition sparse and dense ECE are 0.1199 ± 0.0247 and 0.0158 ± 0.0032 (ΔECE = 0.1041 ± 0.0215). ΔECE* = 0.0068 ± 0.0207, positive on fold 1 and fold 4 and negative on fold 0 and folds 2–3. The matched gap is 0.0581 ± 0.0230. The fold-0 gap is 0.0365, with interval 0.0233 to 0.0878. The mask, the checkpoint, and the window were introduced together, so this is not a mask-only effect. simpleKT is sensitivity only and is not in Table 5.

Junyi Academy has no sparse group. Its DKT medium-versus-dense contrast uses the same checkpoint and window, but it is post-hoc. The four-partition means are ECE 0.0345 ± 0.0080 on medium and 0.0043 ± 0.0021 on dense (ΔECE = 0.0302 ± 0.0069; mean count N medium = 3,677). ΔECE* = −0.0032 ± 0.0067. That value is the mean of the per-partition references, not 0.0302 minus the bound at a single N. The matched gap is 0.0115 ± 0.0073. The fold-0 interval is 0.0010 to 0.0434 and applies to that partition. XES3G5M has no model row: check L8 failed (Section III.A), and no corrected model was trained. Our audit of the released question-level sequence files also found 301 training responses repeated at the start of the learner's next window (artifact v4, audits/GATE_B_XES_REPORT.md). Its frequency distribution remains descriptive only.

Table A1 adds group-level AUC and the Brier decomposition (4) under the same protocol. REL was larger on sparse than on dense components: 0.0137 ± 0.0026 versus 0.0001 ± 0.0000 for DKT and 0.0215 ± 0.0076 versus 0.0004 ± 0.0001 for T-KT masked. Sparse-group AUC was 0.7670 ± 0.0188 for DKT and 0.7476 ± 0.0344 for T-KT masked, against overall AUC of 0.7303 and 0.7244 (Table 4). For DKT, sparse-group AUC exceeded overall AUC in every run; for T-KT masked it was lower in folds 1 and 4. Sparse responses were thus ranked about as well as responses overall while their reliability term was larger, which a pooled AUC does not show. REL is binned like ECE and is biased upward in small groups, so it is not a separate test of the sparse–dense difference.

Fig. 3 shows the fold-0 reliability diagrams for the two primary models on ASSISTments 2012. The sparse curves use 430 responses and 15 bins. They are visual checks, not the four-partition estimate, and they do not show that a checkpoint removed or caused the difference. Junyi Academy has no sparse group and XES3G5M is descriptive only, so neither is plotted.

Table 5. Calibration by frequency group under the primary protocol (mean ± SD across four unique partitions; N is the mean count, rounded). ΔECE is (5), ΔECE* is the bias reference (7), and the matched gap is G (9); the 95% confidence interval (CI) is the fold-0 learner-cluster interval of G. Support: Limited if 100 ≤ N < 1,000; Reliable if N ≥ 1,000. Junyi Academy is post-hoc, and its medium group takes the place of the sparse group.

Dataset	Model	Group (support)	N	ECE	ΔECE	ΔECE*	Matched gap	95% CI (fold 0 gap)
ASSISTments	DKT	sparse (Limited)	398	0.0982 ± 0.0093	0.0893 ± 0.0091	−0.0081 ± 0.0122	0.0400 ± 0.0107	0.0170 to 0.0821
ASSISTments	DKT	dense (Reliable)	517,330	0.0089 ± 0.0013	same contrast	same reference	same gap	same interval
ASSISTments	T-KT masked	sparse (Limited)	398	0.1199 ± 0.0247	0.1041 ± 0.0215	0.0068 ± 0.0207	0.0581 ± 0.0230	0.0233 to 0.0878
ASSISTments	T-KT masked	dense (Reliable)	517,330	0.0158 ± 0.0032	same contrast	same reference	same gap	same interval
Junyi (post-hoc)	DKT	medium (Reliable)	3,677	0.0345 ± 0.0080	0.0302 ± 0.0069	−0.0032 ± 0.0067	0.0115 ± 0.0073	0.0010 to 0.0434
Junyi (post-hoc)	DKT	dense (Reliable)	3,200,666	0.0043 ± 0.0021	same contrast	same reference	same gap	same interval

C. Estimability and cold-start groups

Whether a sparse contrast can be estimated is a property of the counts (Tables 6 and 7; RQ2). ASSISTments 2012 has a nonempty sparse group. Under the primary protocol its mean sparse N is 398, which is Limited, and its dense group is Reliable, so the contrast is eligible. Junyi Academy has no sparse group; the post-hoc contrast is medium versus dense,

and the medium group is Reliable. XES3G5M has low-frequency KC rows (1,969 sparse test rows on average), but the model contrast is not eligible: check L8 failed, and the dataset is descriptive only. Strict cold-start and very-sparse groups are not given a primary ECE.

The structural descriptors in Table 7 characterize the datasets, and none of them is treated as a necessary or sufficient condition for the direction of a contrast. Frequency and difficulty were negatively correlated on ASSISTments 2012 ($\rho = -0.241$, with sparse KCs harder) and Junyi Academy ($\rho = -0.416$) but weakly positively correlated on XES3G5M ($\rho = 0.110$).

Table 6. Estimability and evidence status by dataset under the primary protocol. Checks are those of Table 3. Direction refers to ΔECE (5), the size-matched gap G (9), and the bias reference ΔECE* (7); for Junyi Academy the medium group takes the place of the sparse group. Counts and descriptors are in Table 7.

Question	ASSISTments 2012	Junyi Academy	XES3G5M
Sparse group present?	Yes (Limited)	No	KC-level tail only
L1–L8 PASS?	L1–L7 PASS; L8 not applicable (one row per response)	L1–L7 PASS; L8 not applicable (one row per response)	L1–L7 PASS; L8 FAIL
Eligible model contrast?	Yes: DKT and T-KT masked	Post-hoc medium versus dense (DKT)	No
Direction	Mean ΔECE and G positive for both models; G positive in every partition; ΔECE* mixed in sign	Mean ΔECE 0.0302 and G 0.0115 positive; ΔECE* −0.0032	Not evaluated
Uncertainty	Fold-0 learner-cluster interval of G excludes 0 (primary); per-partition intervals in Table A5 also exclude 0	Fold-0 interval of G 0.0010 to 0.0434; post-hoc	Not evaluated
Protocol/evidence status	Primary protocol; primary evidence	Primary-protocol rules; post-hoc, not pre-specified	Descriptive only

Table 7. Counts and structural descriptors by dataset. Train-derived rows use the fold-0 training split and the KCs present in it; test-derived rows are mean counts over the four unique partitions of the split files (KC-level rows for XES3G5M), except the primary-protocol sparse N, which counts nonempty windows as in Table 5. ρ is Spearman's correlation with log(1 + ftrain). Support: R, Reliable; L, Limited; I, Insufficient. XES3G5M is descriptive only.

Quantity	ASSISTments 2012	Junyi Academy	XES3G5M
Train-derived (fold 0)			
KCs present in training	265	1,326	854
KCs with ftrain < 100 (E1)	18.9%	0.0%	21.5%
KCs very sparse / sparse / medium / dense	11 / 39 / 68 / 147	0 / 0 / 33 / 1,293	62 / 122 / 172 / 498
Strict cold-start KCs in fold-0 test	0	0	3
Median ftrain per KC	988	7,074	1,002
Test-derived (mean over four partitions)			
Sparse-group test rows (E2)	415 (L)	empty	1,969 (R)
Very-sparse test rows	17 (I)	empty	114 (L)
Strict cold-start test rows	1 (I; nonempty in 1 of 4)	empty	5 (I; nonempty in 4 of 4)
Mean sparse N, primary protocol	398 (L)	No sparse group	Not evaluated (descriptive only)
Structural descriptors (fold-0 training)			
Frequency–difficulty ρ	−0.241	−0.416	+0.110
Median item support	32 (dense 198; sparse 1)	18 (dense 18; sparse –)	3 (dense 9; sparse 1)
Frequency–position ρ	−0.326	−0.324	−0.126

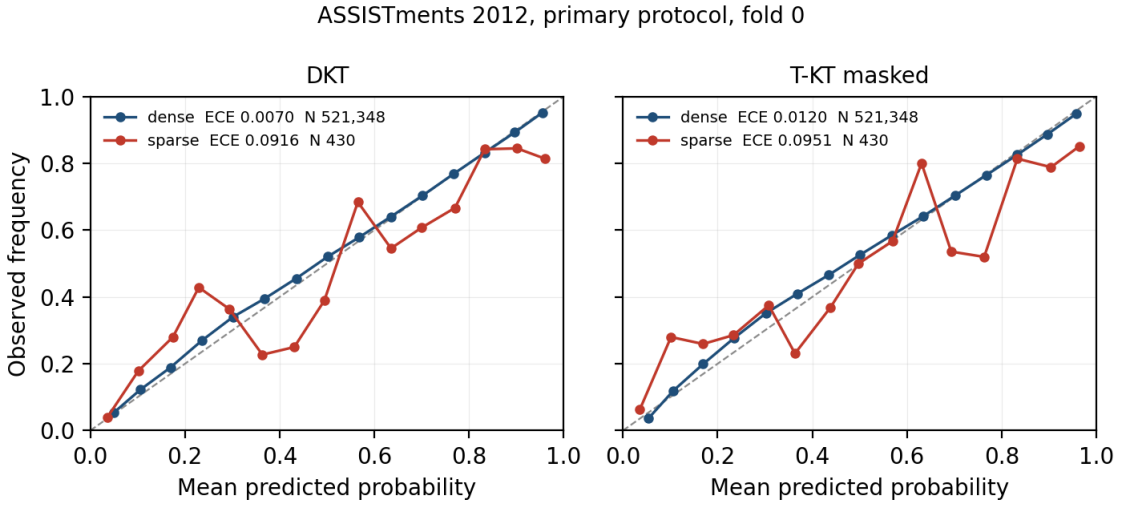


Fig. 3. ASSISTments 2012, primary protocol, fold 0: reliability diagrams for DKT (left) and T-KT masked (right), 15 equal-width bins, nonempty windows. Sparse group N = 430 (Limited); dense group N = 521,348 (Reliable).

D. Sensitivity to checkpoint and test window

Table A2 is the ASSISTments DKT grid of checkpoint and test window; every cell is ΔECE, and no cell is missing. At a fixed test window, the validation-AUC checkpoint had a lower ΔECE than the final checkpoint in all five runs. This checkpoint-associated difference ranged from 0.0181 to 0.0774 (0.0217 to 0.0774 with windows of 200 and 0.0181 to 0.0729 with the full history). At a fixed checkpoint, the window-associated change (window 200 minus full history) ranged from −0.0131 to +0.0068 and had no consistent sign. The two runs that share one learner partition (folds 2 and 3) differ by up to 0.0268 in the final/full cell, which exceeds the smallest checkpoint-associated differences. The direction of the checkpoint-associated difference was therefore consistent across the reported runs, but its magnitude comes from one model on one dataset and is not a general causal estimate. Final × full versus validation × window 200 changes both factors and is not assigned to either alone. Raw ΔECE stays positive in every cell; ΔECE* in the primary cell is positive only in fold 1. Alternative frequency cut points were not evaluated under the primary protocol.

E. Threshold analysis

At $\tau = 0.7$ on fold 0 (Table A3), DKT advanced 200 of 430 sparse responses and 300,610 of 521,348 dense responses. FAR was 0.215 on sparse and 0.179 on dense components, so ΔFAR = 0.036, with a KC-clustered 95% interval of −0.041 to 0.125. For T-KT masked, FAR was 0.232 and 0.182, and ΔFAR = 0.051 (−0.019 to 0.141). Both fold-0 intervals for ΔFAR include 0. Observed FAR exceeded the model's own expected FAR on sparse components by 0.073 for DKT and 0.113 for T-KT masked, against 0.001 and 0.006 on dense components. Across the four unique partitions (Table A4), mean ΔFAR was 0.021 ± 0.019 for DKT and 0.036 ± 0.038 for T-KT masked, and it was positive in 3 of the 4 partitions for DKT and 3 of the 4 for T-KT masked. The sparse advance sets are small, so these figures describe one threshold on these partitions. They are not evidence that sparse advances fail more often in general.

V. DISCUSSION

A. What the results show

The results support a limited claim. Overall AUC can hide frequency-stratified calibration differences. Both primary models showed the difference; its observed magnitude differed between them and across partitions, checkpoints, and datasets. Where a sign varied, it was the sign of the bias reference ΔECE*, not the presence of the calibration difference. On ASSISTments 2012, DKT and T-KT masked have a positive raw ΔECE and a positive size-matched gap. The fold-0 interval for that gap excludes 0. ΔECE* does not exclude a small-sample explanation on every partition, and it is not a significance test. The fold-0 interval is not evidence that all four partitions are significant. Junyi Academy has no sparse group. Its post-hoc ΔECE* is −0.0032. XES3G5M is a descriptive audit only. The study does not claim that a validation checkpoint removes sparse miscalibration, and it does not claim that the final checkpoint causes it.

B. Why the datasets differ

The datasets differ first in whether a sparse contrast exists (Table 6). Junyi Academy's exercise-level components all have at least 100 training responses under this split, so the sparse group is empty. ASSISTments 2012 has a Limited sparse group and is the only dataset with a primary model contrast. XES3G5M can describe frequency and estimability, but its model scores were produced by a route that fails L8. Those scores are excluded. The two primary models showed the same direction with different observed magnitudes. simpleKT was not a primary comparison. The study does not identify a mechanism that would make rare practice itself the cause of miscalibration.

C. Implications for adaptive learning platforms

For platforms that use KT probabilities to skip practice, assign remediation, or declare mastery, the results point to four checks before a single global threshold is trusted. (a) Occupancy: count KCs and test responses by training frequency in the split actually used, and report empty or low-support groups. (b) Calibration: report ECE and the Brier decomposition by group alongside AUC, and compare small-group ECE with the bias bound (6). (c) Decision error: if a fixed threshold is evaluated, report FAR and FPR together with their denominators. (d) Leakage and alignment: verify the checks in Table 3 in training as well as in scoring; when a response carries several KCs, construct the model input so that no KC row of the response reveals its outcome to another; in attention-based models, mask the target and later responses during training; and match each exported prediction to the right learner, item, KC, and sequence position.

Post-hoc calibration methods such as temperature scaling [17] and Platt scaling [31] were outside our scope; whether they close the gap on sparse KCs remains open.

D. Limitations

Several limits qualify these findings. The label is the next response, not a mastery decision, and no classroom trial was run. The primary contrast is one dataset, ASSISTments 2012, with a Limited sparse group. Four unique partitions describe variability. They are not four independent population samples, and fold 2 is the same learners as fold 3. ΔECE* is a bias reference. The primary matched interval covers fold 0 only; the Table A5 intervals for the other partitions are supplementary. T-KT masked changed the mask, the checkpoint, and the window together. Junyi DKT is post-hoc. Junyi T-KT was not retrained with a mask. XES3G5M is a descriptive audit only. BKT was not scored. FAR uses one threshold, and its interval covers fold 0 only. None of the analyses identifies a causal effect of training frequency on calibration.

VI. CONCLUSION

Overall AUC can hide frequency-stratified calibration differences. The observed magnitude differed between the two primary models and across checkpoints, partitions, and

datasets; these are descriptive differences, not designed comparisons. On ASSISTments 2012, under the primary protocol, DKT and a causally masked Transformer show a higher sparse ECE than dense ECE. The fold-0 size-matched interval excludes 0. The four-partition bias reference ΔECE* is −0.0081 for DKT and 0.0068 for the masked Transformer, and that reference is not a significance test. Junyi Academy has no sparse group; its post-hoc ΔECE* is −0.0032. XES3G5M is a descriptive audit only, with no model-based result. The study does not recommend a new tracing model or a classroom policy. It asks evaluations that feed instructional rules to report calibration and support by training frequency, and to keep the bias reference separate from a size-matched interval.

APPENDIX

Table A1. Group-level discrimination and Brier decomposition under the primary protocol (mean ± SD across four unique partitions; 15 equal-width bins; nonempty windows). N is the mean count, rounded. The base-rate reference predicts each partition's training proportion correct, so its AUC is 0.5000 and its RES is 0 by construction. simpleKT is sensitivity; Junyi is post-hoc and has no sparse group. REL and RES are binned and biased upward in small groups.

Dataset	Model	Group	N	AUC	ACC	Brier	UNC	REL	RES
ASSISTments 2012	DKT	sparse	398	0.7670 ± 0.0188	0.7318 ± 0.0132	0.1904 ± 0.0088	0.2356 ± 0.0119	0.0137 ± 0.0026	0.0589 ± 0.0090
ASSISTments 2012	DKT	medium	5,867	0.8131 ± 0.0125	0.7342 ± 0.0132	0.1758 ± 0.0069	0.2484 ± 0.0013	0.0013 ± 0.0005	0.0733 ± 0.0055
ASSISTments 2012	DKT	dense	517,330	0.7285 ± 0.0004	0.7354 ± 0.0010	0.1797 ± 0.0004	0.2100 ± 0.0003	0.0001 ± 0.0000	0.0301 ± 0.0001
ASSISTments 2012	T-KT masked	sparse	398	0.7476 ± 0.0344	0.7198 ± 0.0201	0.2039 ± 0.0187	0.2356 ± 0.0119	0.0215 ± 0.0076	0.0530 ± 0.0125
ASSISTments 2012	T-KT masked	medium	5,867	0.8042 ± 0.0114	0.7243 ± 0.0104	0.1820 ± 0.0059	0.2484 ± 0.0013	0.0036 ± 0.0005	0.0695 ± 0.0046
ASSISTments 2012	T-KT masked	dense	517,330	0.7227 ± 0.0006	0.7315 ± 0.0007	0.1817 ± 0.0004	0.2100 ± 0.0003	0.0004 ± 0.0001	0.0284 ± 0.0001
ASSISTments 2012	Base rate	sparse	398	0.5000	0.6101 ± 0.0556	0.2453 ± 0.0218	0.2356 ± 0.0119	0.0097 ± 0.0100	0.0000 ± 0.0000
ASSISTments 2012	Base rate	medium	5,867	0.5000	0.5363 ± 0.0198	0.2742 ± 0.0078	0.2484 ± 0.0013	0.0258 ± 0.0065	0.0000 ± 0.0000
ASSISTments 2012	Base rate	dense	517,330	0.5000	0.6999 ± 0.0007	0.2100 ± 0.0003	0.2100 ± 0.0003	0.0000 ± 0.0000	0.0000 ± 0.0000
ASSISTments 2012	simpleKT (sensitivity)	sparse	398	0.7850 ± 0.0362	0.7363 ± 0.0180	0.1798 ± 0.0112	0.2356 ± 0.0119	0.0128 ± 0.0032	0.0682 ± 0.0152
ASSISTments 2012	simpleKT (sensitivity)	medium	5,867	0.8262 ± 0.0106	0.7461 ± 0.0117	0.1700 ± 0.0060	0.2484 ± 0.0013	0.0013 ± 0.0004	0.0793 ± 0.0045
ASSISTments 2012	simpleKT (sensitivity)	dense	517,330	0.7698 ± 0.0014	0.7534 ± 0.0015	0.1686 ± 0.0006	0.2100 ± 0.0003	0.0009 ± 0.0002	0.0420 ± 0.0004
Junyi (post-hoc)	DKT	medium	3,677	0.6998 ± 0.0167	0.6564 ± 0.0147	0.2152 ± 0.0035	0.2432 ± 0.0013	0.0020 ± 0.0009	0.0299 ± 0.0043
Junyi (post-hoc)	DKT	dense	3,200,666	0.7554 ± 0.0012	0.7470 ± 0.0013	0.1714 ± 0.0006	0.2078 ± 0.0007	0.0000 ± 0.0000	0.0360 ± 0.0004
Junyi (post-hoc)	Base rate	medium	3,677	0.5000	0.5822 ± 0.0082	0.2578 ± 0.0033	0.2432 ± 0.0013	0.0146 ± 0.0020	0.0000 ± 0.0000
Junyi (post-hoc)	Base rate	dense	3,200,666	0.5000	0.7055 ± 0.0018	0.2078 ± 0.0007	0.2078 ± 0.0007	0.0000 ± 0.0000	0.0000 ± 0.0000

Table A2. ASSISTments DKT sensitivity grid. Each main cell is ΔECE = ECE sparse − ECE dense. best/200 is the primary protocol. ΔECE* in the extra column is the bias reference for that primary cell, not a confidence interval. Fold 2 and fold 3 share learners. No cell is missing. This table does not attribute the final/full versus best/200 gap to one factor.

Fold	final/full	final/200	best/full	best/200	ΔECE* best/200	N sparse best/200
fold 0	0.1689	0.1620	0.0977	0.0846	-0.0088	430
fold 1	0.1746	0.1732	0.1017	0.1016	0.0074	423
fold 2	0.1086	0.1084	0.0905	0.0867	-0.0118	386
fold 3	0.1354	0.1316	0.0867	0.0935	-0.0050	386
fold 4	0.1319	0.1335	0.0822	0.0808	-0.0224	352

Table A3. Threshold analysis at τ = 0.7 under the primary protocol (ASSISTments 2012, fold 0, nonempty windows). FAR intervals are KC-clustered percentile bootstrap intervals (B = 2,000; seed 20261021). ΔFAR is 0.036 [−0.041, 0.125] for DKT and 0.051 [−0.019, 0.141] for T-KT masked. Excess is FAR minus E[FAR].

Model	Group	N	N_{adv}	N_{inc}	FAR [95% CI]	E[FAR]	Excess	FPR
DKT	sparse	430	200	195	0.215 [0.138–0.304]	0.142	0.073	0.221
DKT	dense	521,348	300,610	156,258	0.179 [0.170–0.187]	0.178	0.001	0.344
T-KT masked	sparse	430	198	195	0.232 [0.163–0.320]	0.119	0.113	0.236
T-KT masked	dense	521,348	298,201	156,258	0.182 [0.173–0.191]	0.176	0.006	0.347

Table A4. ΔFAR at τ = 0.7 across four unique partitions under the primary protocol (ASSISTments 2012; fold 2 and fold 3 averaged first; the SD is descriptive). N, Nadv, and Ninc are sparse-group mean counts, rounded.

Model	Mean ΔFAR	SD	Range	Partitions with ΔFAR > 0	Mean N	Mean N_{adv}	Mean N_{inc}
DKT	0.021	0.019	−0.001 to 0.038	3/4	398	191	157
T-KT masked	0.036	0.038	−0.007 to 0.081	3/4	398	192	157

Table A5. Size-matched gap G (9) and its learner-cluster 95% interval by unique partition (ASSISTments 2012, primary protocol; B = 2,000, inner draws 200). Fold 0 is the primary uncertainty analysis. The other intervals are supplementary, use a generator seeded [20261021, f] for the first fold f of the partition, and are not combined into one significance claim. For folds 2–3, each replicate resamples the shared learners once and averages G over the two runs. The mean ± SD rows reproduce Table 5.

Model	Partition	N sparse	G	95% interval	Role
DKT	fold 0	430	0.0351	0.0170 to 0.0821	Primary
DKT	fold 1	423	0.0546	0.0331 to 0.1040	Supplementary
DKT	folds 2–3	386	0.0406	0.0256 to 0.0889	Supplementary
DKT	fold 4	352	0.0295	0.0089 to 0.0848	Supplementary
DKT	Mean ± SD		0.0400 ± 0.0107		Table 5
T-KT masked	fold 0	430	0.0365	0.0233 to 0.0878	Primary
T-KT masked	fold 1	423	0.0834	0.0513 to 0.1439	Supplementary
T-KT masked	folds 2–3	386	0.0408	0.0286 to 0.0907	Supplementary
T-KT masked	fold 4	352	0.0715	0.0450 to 0.1251	Supplementary
T-KT masked	Mean ± SD		0.0581 ± 0.0230		Table 5

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

K.-T.N.: Conceptualization, Methodology, Software, Formal analysis, and Writing (led diagnostic design, experiments, and manuscript). T.D.M. and D.N.T.: Software (data processing and baseline runs). C.T.N.: Methodology (methodological review). V.-H.N.: Supervision and Writing (manuscript revision). All authors approved the final version.

ETHICAL STATEMENT

This study is a secondary analysis of three publicly released educational datasets (ASSISTments 2012, Junyi Academy, and XES3G5M) that the providers de-identified before release. We did not recruit participants, contact learners, parents, guardians, or teachers, or conduct any classroom intervention, and we collected no new consent forms, logs, recordings, or personal data. No direct identifiers were accessed, and no attempt was made to re-identify any learner. The data were used under the providers' terms; Junyi Academy releases its data under CC BY-NC-SA 4.0, and XES3G5M is released under the MIT license. The study was neither commissioned by nor conducted with an educational technology company. Under the research policy of Hung Yen University of Technology and Engineering, a secondary analysis of publicly released, de-identified data does not require prospective human-subjects review, so no ethics protocol number applies. No exemption from data-protection law is claimed.

DATA AND CODE AVAILABILITY

Artifact v4 contains the code snapshot at commit 4ff6683, the configuration files from that commit, the five learner-partition seeds and the bootstrap seed, the per-run training logs, the DKT token-0 log, and the masked T-KT log for unit tests UT-A1 through UT-A7. It also contains the prediction exports for folds 0-4 of ASSISTments DKT on the four Table A2 cells, ASSISTments T-KT masked, ASSISTments simpleKT, and Junyi DKT, and the per-partition base-rate constants used in Table A1. From an extracted copy, one command recomputes the supported cells of Tables 4, 5, A1, A2, A3, A4, and A5, the size-matched gaps, and the fold-0 intervals from those exports without retraining; a second command recomputes the supplementary intervals of Table A5. The published gap uses a second random sample of exactly N rows. The size-match function inside the 4ff6683 snapshot keeps a prefix and does not reproduce that gap. The XES3G5M audit files cited in Sections III.A and IV.B are included. The artifact does not

contain the raw datasets or the processed split files (Tables 6 and 7 are included as recorded outputs), the per-partition records of checks L1–L7 behind Table 3 (only the runner code is included), the model weights, the prediction exports of one additional run (labelled fold 5 in the artifact, added after the analysis plan; its training logs are included) that this paper does not report, the slide-window exports, Junyi simpleKT, XES3G5M predictions, or unmasked T-KT. XES3G5M stays not evaluated, and retraining is not supported. The artifact is available from the corresponding author on request, and a public repository will be released upon acceptance.

GENERATIVE AI STATEMENT

During manuscript preparation, the authors used Cursor 3.18.9 (with Grok 4.6 and Grok 4.7) for language polishing, formatting, consistency checking, and preparation of reproducibility prompts; ChatGPT (OpenAI; GPT-5.6) to edit an intermediate revision and to compare revised versions; and Claude (Anthropic; Claude Opus 5.5, and Claude Sonnet 5.5 inside Cursor) to review the manuscript and the analysis code, to draft structural and language revisions, and, in the final revision, to write and run scripts that recompute the reported tables from the prediction files. No AI tool trained a model or changed a prediction file. The authors checked every AI-assisted passage, script, and recomputed value against the underlying code and prediction files, revised it, and take full responsibility for the content. No AI tool is listed as an author.

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